

1 Scalar Diffraction

Learning outcomes: *In this unit we will use the Huygens-Fresnel construction for solving scalar diffraction problems. We will derive the Fraunhofer diffraction equation and give conditions under which the Fraunhofer equation is appropriate in computing solutions to diffraction problems. We will explore the origin of the link between Fourier theory and Fraunhofer diffraction. We will use the three different variables used to describe Fraunhofer diffraction. We will explain the difference between the observed diffraction intensity pattern and the diffracted field.*

Contents

1	Scalar Diffraction	1
1.1	Introduction to Physical Optics	1
1.2	Fraunhofer or Far Field Diffraction	2
1.2.1	Huygens-Fresnel Theory	2
1.2.2	Harmonic Spherical Waves	2
1.2.3	The Obliquity Factor	2
1.3	General Diffraction Problems	3
1.4	Fraunhofer or Far-Field Diffraction	4
1.5	Derivation of the Fraunhofer Diffraction Equation	4
1.5.1	Simplifying the Equation	6
1.5.2	Expressing $\mathbf{R}$ in Terms of $(\mathbf{x}, \mathbf{y})$ and (ζ, η)	7
1.5.3	The Fraunhofer Condition	7
1.5.4	The Fraunhofer Diffraction Equation	8
1.6	Observed Diffraction Patterns and Intensity or Irradiance	8
1.7	Diffraction Space Variables	9
1.8	Problems	10
1.9	Revision and Extra Problems	10

1.1 Introduction to Physical Optics

Light is an electromagnetic wave and as such, its behaviour is determined by Maxwell's equations of electromagnetism. In principle, all optical phenomena can be reduced to solving Maxwell's equations for a given set of boundary conditions. Exact solutions gained in this way are rare and usually difficult to obtain. The first, rigorous solution was obtained by Sommerfeld in 1896 where he determined the effect of a plane wave encountering a semi-infinite, perfectly conducting screen. There have since been a few other rigorous solutions.

The approach taken in this section will be to use a heuristic method: the Huygens-Fresnel theory. This is a powerful approach from which one can gain insight into physical optical phenomena. A more rigorous link between Maxwell's electromagnetic theory and the Huygens-Fresnel approach will be made when we consider Kirchhoff's diffraction theory.

Scalar diffraction is used to solve problems in Electromagnetism where the vector nature of the E.M. waves can be neglected. That is, problem where the E.M. boundary conditions for the Transverse electric and transverse magnetic waves will yield the same result. Problems where the specific orientation of the $\mathbf{E}$ and $\mathbf{H}$ fields are important include thin film optics, Mie scattering, waveguides, etc.

1.2 Fraunhofer or Far Field Diffraction

1.2.1 Huygens-Fresnel Theory

Huygens developed the idea that light can be considered as a series of wavefronts where each new front creates the next wavefront by emitting a series of secondary disturbances - secondary wavelets - where the new wavefront is the envelop of these wavelets. With this simple construction, Huygens was able to reproduce all of geometric optics - i.e. the laws of reflection and refraction of light.

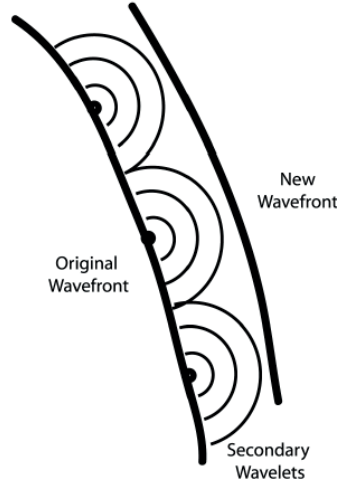


Figure 1: Huygens' secondary wavelet construction. The original wavefront acts as a series of sources for secondary wavelets. The envelop of these secondary wavelets then defines the new wavefront.

Fresnel improved on this idea by adding that the secondary wavelets should interact by mutual interference. With this sophistication, Fresnel was able to use Huygens' idea to account for diffraction phenomena and hence physical optics.

1.2.2 Harmonic Spherical Waves

Waves that are created by a point source will radiate in all directions to produce a spherical wavefront. This can be represented mathematically by:

$$\psi(r, t) = \left(\frac{\mathcal{A}}{r} \right) e^{i(kr - \omega t)} \quad (1)$$

where r is the distance from the point source, t is time, $\mathcal{A}$ is the source strength and $k = 2\pi/\lambda$ is the magnitude of the wave vector, and λ is the wavelength. The wave vector is a measure of the momentum carried by the wave. For diffraction problems, we will only consider the spatial part of the wave and hence drop any explicit time dependence.

1.2.3 The Obliquity Factor

There is an obvious problem with the Huygens construction: the wavefront will generate a backward moving wave as well as the new forward moving wavefront. Fresnel solved this problem by adding a term which biased against the backward moving wave. This term, $\kappa(\chi)$, is called the obliquity or inclination factor and it depends on the angle, χ , between the primary wave direction and the secondary wave direction.

For most of this course, the obliquity factor will be assumed to be approximately equal to one. For the case of a diffraction aperture illuminated by plane wave, the obliquity factor used by Fresnel takes on the form

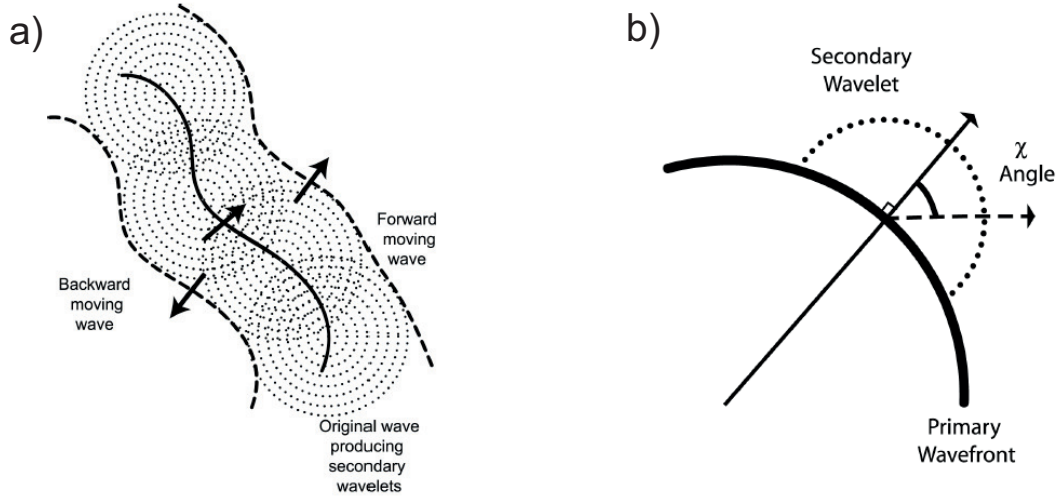


Figure 2: a) The backward moving wave. The Huygens' construction results in a backward moving wave produced by the envelop of the secondary wavelets. This was corrected by Fresnel's inclusion of an obliquity or inclination factor. b) The obliquity factor. The amplitude of the secondary wavelet is modulated by a factor, $\kappa(\chi)$, which is a function of the angle, χ , between the primary wave direction and the direction of an element of the secondary wavelet.

$$\kappa(\chi) = \frac{1 + \cos(\chi)}{2} \quad (2)$$

This equation has the property that there is no backward wave. The general form of the obliquity factor is derived using Kirchhoff's scalar theory.

Thus the secondary wavelet will have the form,

$$\psi(r, t) = \left(\frac{\mathcal{A}}{r} \right) e^{i(kr - \omega t)} \kappa(\chi) \quad (3)$$

1.3 General Diffraction Problems

The general diffraction problem has three components: a source of light; an observation plane where one wishes to compute the light wave; and an intervening aperture screen or obstacle which alters the passage of the light from the source to the screen.

The general strategy that we will use to solve this problem will be:

1. Determine the source strength, $\mathcal{A}$, for all points in the source.
2. Pick a point (x, y) in the aperture/obstacle plane and determine the field at (x, y) just before the obstacle by adding up contributions from all parts of the source.
3. Characterise the effect of the aperture/obstacle on the incident wave by an aperture function $f(x, y)$ where the field emerging from the obstacle is given by the product of the incident wave, $\psi(x, y)$ with the aperture function $f(x, y)$.
4. Use the field emerging from the aperture/obstacle, $\psi(x, y) \cdot f(x, y)$ as a new source.
5. Pick a point (ζ, η) in the observation screen and add up all the contributions arriving at (ζ, η) from the radiation leaving the aperture/obstacle.
6. Repeat the procedure for all points (ζ, η) in the observation screen.

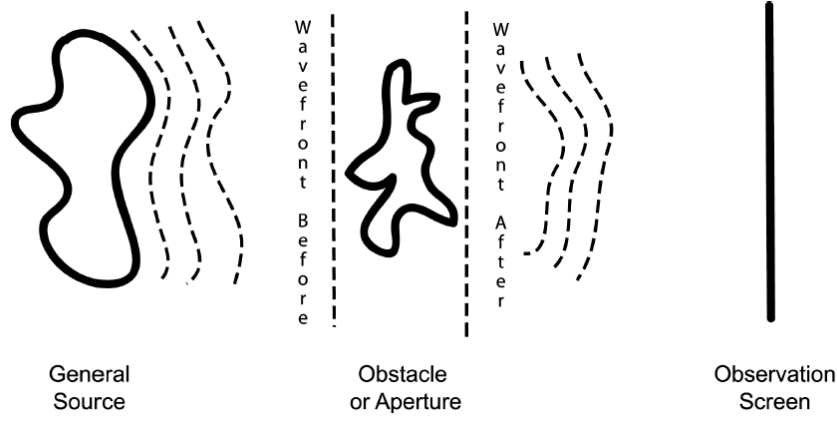


Figure 3: General diffraction problem. A source gives rise to optical radiation which illuminates an aperture or an obstacle. This, in turn, modifies the wavefront which then illuminates an observation screen.

1.4 Fraunhofer or Far-Field Diffraction

Fraunhofer diffraction refers to diffraction phenomena that are observed at large distances from the diffracting object, hence it is often referred to as “far-field diffraction”. The conditions for Fraunhofer diffraction are, in general terms:

1. illuminating light is coherent and can be considered a surface of constant phase,
2. the object/aperture is small compared to the distance between the object and the observation plane.

The first condition can be satisfied by using a distant point source or by illuminating the object with plane wave radiation.

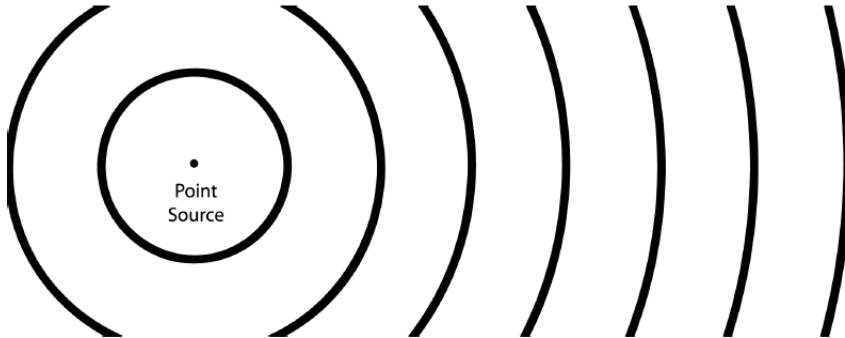


Figure 4: Distant point source produces plane waves.

The second condition will be used for several important approximations during the derivation of the Fraunhofer equation.

1.5 Derivation of the Fraunhofer Diffraction Equation

Our aim is to derive the form of the Fraunhofer or far-field diffraction equation. The geometry needed is described in Figure. For simplicity, we assume that the diffraction aperture is illuminated by a uniform, coherent monochromatic plane wave. This assumption is unnecessary but it simplifies the algebra. The relaxation of this assumption is left as an exercise

The effect of the aperture on the incoming wave is described by a multiplicative aperture function, $f(x, y)$ which is complex. The amplitude of $f(x, y)$ represents absorption of light while the phase

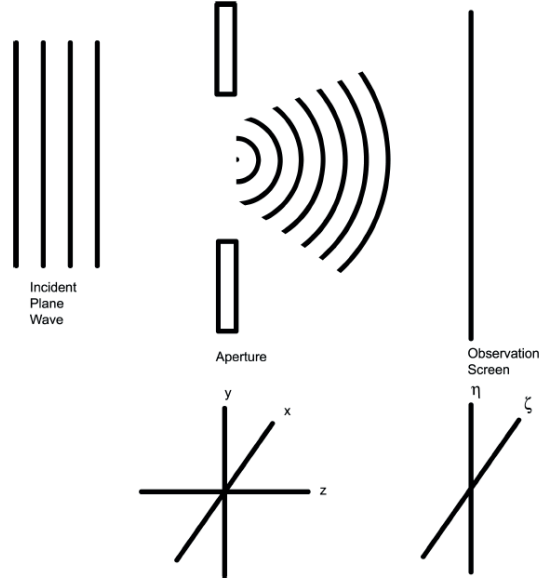


Figure 5: Diffraction geometry and coordinate systems. The upper portion of the Figure shows the geometry: an incident plane wave illuminates an aperture. The aperture alters the wave and the transmitted wavefront acts as a source of secondary wavelets. These wavelets interfere to produce an intensity pattern at an observation screen. The bottom portion of the Figure shows the coordinate systems: (x, y, z) in the aperture plane; and (ζ, η) in the observation plane.

component reflects any changes the aperture produces in the phase of the wavefront. For example, for a simple mask, the aperture function will have a value of zero where the mask is opaque and unity where the mask is transparent. If the aperture is a greyscale transparency, the aperture function will vary continuously between zero and unity depending on the transparency. Finally, if the aperture is transparent, but consists of a refractive material of varying thickness, then the aperture function will be complex, with an amplitude equal to one where the exponent will describe the phase shifts induced by the refractive material.

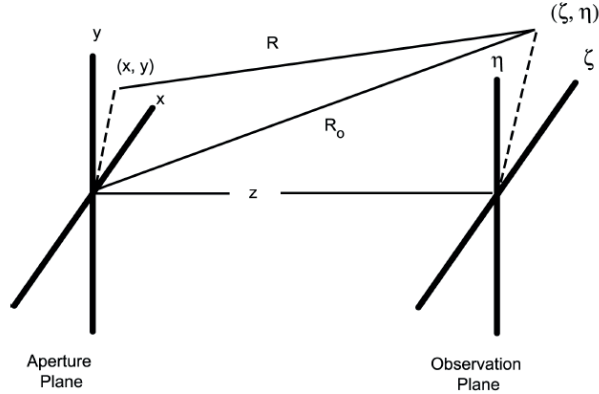


Figure 6: Aperture and observation planes. The Figure shows the geometry used in deriving the form of the Fraunhofer diffraction equation.

A point (x, y) in the aperture plane will act as the source of secondary wavelets of the form,

$$\psi(x, y, r, \chi) = f(x, y) \frac{e^{ikr}}{r} \kappa(\chi). \quad (4)$$

The field arriving at a point (ζ, η) in the observation plane can be computed by adding up all secondary wavelet contributions arriving at (ζ, η) . Here $\mathcal{A}$ is assumed to be unity. This results in an integral over the aperture plane,

$$\Phi(\zeta, \eta) = \iint_{\text{aperture}} \psi(x, y) dx dy \quad (5)$$

$$= \iint_{\text{aperture}} f(x, y) \frac{e^{ikR}}{R} \kappa(\chi) dx dy. \quad (6)$$

1.5.1 Simplifying the Equation

Although the Fraunhofer diffraction pattern can be computed from Eqn 5, major insight and mathematical power can be gained by simplifying this equation.

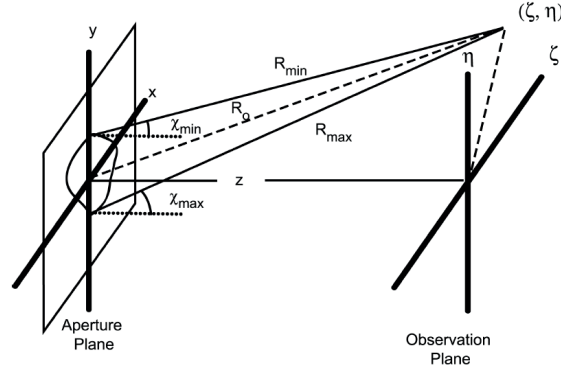


Figure 7: Approximations justified by a small diffraction aperture. The small aperture limits the values of both R and χ for points, (x, y) inside the aperture. The mean values of R and χ will depend on the coordinates of the observation point, (ζ, η) .

The condition that the aperture must be small compared with the distance to the observation screen can be used to simplify the diffraction equation. For a given point (ζ, η) in the observation plane:

$$R \approx R_0 \quad (7)$$

$$\kappa(\chi) = \text{const.} \quad (8)$$

where the constant will depend on the mean value of c for (ζ, η) with respect to the small aperture. Fig. 7 shows how the range of values of both R and χ will be limited by the aperture size. Note, however, that the mean values of both R and χ will critically depend on the coordinates of the observation point, (ζ, η) .

R_0 cannot be substituted for R in the exponent under the integration sign in Eqn. 5 since the exponent is,

$$ikR = 2\pi i \left(\frac{R}{\lambda} \right) \quad (9)$$

which would require that the difference between R and R_0 must be very small compared with the wavelength of light before substitution was acceptable.

$$R - R_0 \ll \lambda \quad (10)$$

Using the above approximations Eqn. 5, the equation for the total diffracted field at (ζ, η) becomes,

$$\Phi(\zeta, \eta) = \frac{\kappa(\chi(\zeta, \eta))}{R(\zeta, \eta)} \iint_{\text{aperture}} f(x, y) e^{ikR} dx dy. \quad (11)$$

1.5.2 Expressing R in Terms of (x, y) and (ζ, η)

To simplify the exponent of Equation 1.9, R must first be expressed in terms of the coordinates in both the aperture and observation planes, (x, y) and (ζ, η) , respectively. From the geometry of the problem and Pythagoras' theorem, the distances, R and R_0 , can be expressed as,

$$R^2 = (\zeta - x)^2 + (\eta - y)^2 + z^2 \quad (12)$$

$$R_0^2 = \zeta^2 + \eta^2 + z^2 \quad (13)$$

hence,

$$R^2 = R_0^2 - 2(\zeta x + \eta y) + (x^2 + y^2) \quad (14)$$

$$= R_0^2 \left\{ 1 + \frac{-2(\zeta x + \eta y) + (x^2 + y^2)}{R_0^2} \right\} \quad (15)$$

R can be evaluated using the Taylor expansion,

$$(1 + x)^{\frac{1}{2}} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \dots \quad (16)$$

Applying this expansion we get

$$R = R_0 - \frac{(\zeta x + \eta y)}{R_0} + \frac{(x^2 + y^2)}{2R_0} \quad (17)$$

1.5.3 The Fraunhofer Condition

The Taylor series expansion serves two purposes. First, it will be used to simplify the diffraction Eqn. Second, it will mathematically define what we mean by an aperture being “small” with respect to the distance to the observation screen.

The total diffracted wave at (ζ, η) is now given by

$$\Phi(\zeta, \eta) = \frac{\kappa(\chi) e^{ikR_0}}{R_0} \iint_{\text{aperture}} f(x, y) e^{ik\alpha(x, y, \zeta, \eta)} dx dy. \quad (18)$$

where

$$\alpha(x, y, \zeta, \eta) = -\frac{(\zeta x + \eta y)}{R_0} + \frac{(x^2 + y^2)}{2R_0} \quad (19)$$

The phase of the exponent under the integration sign is now given by,

$$\text{phase} = k\alpha(x, y, \zeta, \eta) \quad (20)$$

$$= 2\pi \left(-\frac{(\zeta x + \eta y)}{\lambda R_0} + \frac{(x^2 + y^2)}{2\lambda R_0} \right) \quad (21)$$

The condition on the aperture being small compared to the distance to the observation screen can be used to eliminate all terms greater than first order in (x, y) if

$$\frac{(x^2 + y^2)}{2\lambda R_0} \ll 1 \quad (22)$$

Eqn 22 says that all points inside the aperture must be such that the term on the left hand side of the equation is very small compared to one. Thus, the equation defines the maximal dimensions of the aperture. Eqn 22 is called the Fraunhofer condition and defines how small an aperture must be for the Fraunhofer diffraction equation to be valid.

1.5.4 The Fraunhofer Diffraction Equation

With the Fraunhofer condition, Eqn. 22, the expression for the diffracted field (Eqn 18) becomes,

$$\Phi(\zeta, \eta) = \frac{\kappa(\chi)e^{ikR_0}}{R_0} \iint_{\text{aperture}} f(x, y) e^{-2\pi i \frac{\zeta x + \eta y}{\lambda R_0}} dx dy. \quad (23)$$

By changing the variables to $s = (u, v)$ where,

$$u = \frac{\sin \theta}{\lambda} \approx \frac{1}{\lambda} \frac{\zeta}{R_0} \quad (24)$$

$$v = \frac{\sin \phi}{\lambda} \approx \frac{1}{\lambda} \frac{\eta}{R_0} \quad (25)$$

the diffracted field is given by,

$$\Phi(u, v) = \frac{\kappa(\chi)e^{ikR_0}}{R_0} \iint_{\text{aperture}} f(x, y) e^{-2\pi i(xu + yv)} dx dy. \quad (26)$$

But because there is a limit on the size of the diffraction aperture by the Fraunhofer condition (Eqn. 22), the aperture function $f(x, y)$ will be zero when,

$$x^2 + y^2 > r_{max}^2 \quad (27)$$

where r_{max} is the limiting dimension of the aperture as defined by the Fraunhofer condition. This allows us to extend the limits of integration to $\pm\infty$. The diffracted field now becomes,

$$\Phi(u, v) = \frac{\kappa(\chi)e^{ikR_0}}{R_0} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-2\pi i(xu + yv)} dx dy, \quad (28)$$

$$= \frac{\kappa(\chi)e^{ikR_0}}{R_0} \cdot \mathcal{F} \{f(x, y)\} \quad (29)$$

1.6 Observed Diffraction Patterns and Intensity or Irradiance

For electromagnetic radiation in the optical wavelength range, the diffraction pattern observed on a screen or recorded by a detector represents the intensity or irradiance distribution produced by the diffracted field. The intensity is a measure of the energy in the light field and it is determined from the field by,

$$\text{Intensity} = \Phi \cdot \Phi^* \quad (30)$$

where the asterisk indicates the complex conjugate. The intensity or irradiance is real. Thus, the intensity distribution produced by Fraunhofer diffraction is given by,

$$I(u, v) = \Phi(\zeta, \eta) \cdot \Phi(\zeta, \eta)^* \quad (31)$$

$$= \left(\frac{\kappa(\chi)}{R_0} \right)^2 \mathcal{F} \{f(x, y)\} \mathcal{F}^* \{f(x, y)\} \quad (32)$$

Many misconceptions about Fraunhofer diffraction arise from a failure to compute the diffraction intensity distribution rather than just the diffracted field. Note that it is possible to directly measure the electromagnetic field for other types of radiation such as radio waves.

1.7 Diffraction Space Variables

Several different variables will be used in this course to describe diffraction phenomena. From Fourier theory, the most natural variable is the Fourier space variable, $s = (u, v)$. s is in units of inverse meters and will also be referred to as the reciprocal space vector due to its prominent role in solid state physics. s also represents a spatial frequency which is analogous to the usual temporal frequency.

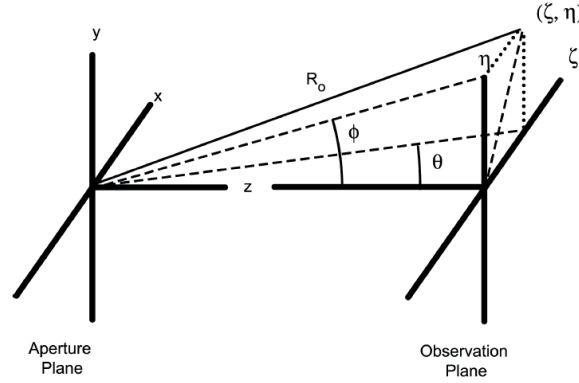


Figure 8: Diffraction variables. The Figure shows the relationship between two different sets of variables that are used to describe diffraction phenomena. The point on the observation plane can be described by using either the coordinates, (ζ, η) or the diffraction angles, (θ, ϕ) .

In diffraction computations, one usually wants to know where features of the diffraction pattern will appear on some screen or observation plane. In this regard, s can be misleading as it is independent of the frequency of the radiation. s is purely related to the properties of the diffracting object.

For the computation of real diffraction problems, one ultimately wants coordinates in the observation plane, (ζ, η) . As per above, these are related to the Fourier variable, $s = (u, v)$ by,

$$\zeta \approx u\lambda R_0 \quad (33)$$

$$\eta \approx v\lambda R_0 \quad (34)$$

where the approximation holds for small angles.

Given the large distances involved between aperture and observation screen (to ensure that the Fraunhofer condition is met), it is often more convenient to describe diffraction phenomena in terms of the angles in which diffracted light emanates from the object (Fig. 8). The angles corresponding to (u, v) are denoted by (θ, ϕ) where

$$\sin(\theta) = u\lambda \quad (35)$$

$$\sin(\phi) = v\lambda \quad (36)$$

$$(37)$$

1.8 Problems

1. A hole, five centimetres in diameter is drilled into a wall. Outside, the wall is illuminated with monochromatic plane wave radiation at a wavelength of 600nm. How far would an observation plane have to be placed in order to observe Fraunhofer diffraction from the hole?
2. In the demonstration that the far field diffraction pattern is equivalent to a Fourier transform of the aperture function, it was assumed that the object producing the diffraction of light is indeed an aperture. Discuss the differences and any difficulties in deriving this equation if the aperture was replaced by a small object which satisfies the Fraunhofer condition.
3. The assumption that the incident field is produced by a plane wave is not essential to the derivation of the Fraunhofer diffraction equation. Assume instead that the aperture is illuminated by a point source, some distance, so from the origin of the aperture coordinate system. Under these conditions, derive the appropriate form of the far-field diffraction equation. Determine the restrictions on the distance, so, with respect to the size of the aperture for the diffraction equation to result in a Fourier relationship between the aperture function and the diffracted field.

1.9 Revision and Extra Problems

1. Starting with Maxwell's equations show that the electromagnetic plane wave satisfies the wave equation.
2. Under what conditions can Maxwell's equations be substituted for the scalar wave equation?
3. Show that for the scalar wave equation, a time harmonic solution of the form $U(\mathbf{r}, t) = U(\mathbf{r})e^{-i\omega t}$ leads to the Helmholtz equation,

$$\nabla^2 U + k^2 U = 0 \quad (38)$$

where k is equal to the wavenumber.

4. Show that a spherical wave of the form $\psi = \frac{\mathcal{A} e^{ikr}}{r}$ is a solution of the Helmholtz equation.